

YIXUAN ASHER ZHANG

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science (MCS)

Champaign, IL
May 2028 (expected)

University of Nottingham

B.Sc. (Hons) Computer Science, First Class · GPA 3.9/4.00

Ningbo, China
Jun 2026

Honors: Outstanding Graduate (Top 5%, 2026); Dean's Scholarship (Top 10%, 2025); Outstanding Student (Top 5%, 2025); Zhejiang Provincial Government Scholarship (Top 1%, 2024); MCM Honorable Mention (2024).

Selected Coursework with 4.0/4.0: Computer Vision (ranked 1st), Computer Security (ranked 1st), Operating Systems & Concurrency, Algorithms & Data Structures, Artificial Intelligence Methods, Databases and Interfaces.

EXPERIENCE

Ningbo AST Machinery Technology

Software Engineer Intern | GitHub

Ningbo, China
May 2026 – Aug 2026

- Shipped a cross-platform multi-material 3D-printing slicer in C++17, Qt, and OpenGL (STEP/STL assemblies, per-part materials, layer masks, G-code) with CMake builds and an Xvfb end-to-end self-test over the full import-to-G-code path.
- Designed a 51-table MySQL CRM (107 foreign keys, 70 constraints) and 10 Vue/FastAPI modules; built an LLM pipeline that produced 345 faculty profiles and 1,127 deduplicated grant records.

Zhiyang Innovation Technology

Machine Learning Engineer Intern

Shanghai, China
May 2025 – Aug 2025

- Cut Jetson Orin Nano inference latency from 220 ms to 95 ms (2.3×) and model size from 87 MB to 14 MB (6.2×) at under 0.5 mAP loss by exporting to ONNX and deploying INT8 TensorRT FP16 behind FastAPI and Docker.
- Raised mAP@0.5 from 0.74 to 0.79 and recall 6.2 points while cutting false positives 12% on a 7:1 imbalanced defect set; training pipeline for 120K+ images cut label noise 18% and data prep from 3 days to under 1, adding 5.6 mAP@0.5.

Beijing Corefire Technology

Software Engineer Intern

Beijing, China
May 2024 – Jul 2024

- Implemented merchant onboarding and POS/QR query APIs for UMFintech in Spring Cloud and MySQL, plus reports that summarized onboarding volume, terminal activations, and provider profit sharing by day, month, or custom range.

PROJECTS

DualPath-CEM | GitHub · Final-year thesis, PyTorch, privacy ML

Nov 2025 – Jun 2026

- Reached 81.96% mean top-1 on a 44,139-image FaceScrub split (526 identities, 5 seeds), 1.63 points above Noise_ARL+CEM, while raising decoder and GAN reconstruction MSE 75.8% and 74.9%.
- Designed dual-path split inference with a Gaussian-protected 16×16×16 SlotCEM tensor and a 256-D MobileNetV3 token fused by calibrated logits (8 slots of 64-D, 64-entry per-class memory).

Manuscript in preparation: Y. Zhang and J. Ren, "DDP: A Disentangled Dual-Path Privacy-Protection Framework for Collaborative Face Inference," Elsevier Pattern Recognition, 2026.

SkillRouter | GitHub · Python, LLM agents

Sep 2026

- Cut per-query agent context from 14,108 to 2,052 tokens (−85.5%) on a 60-skill catalogue while holding Top-1 routing accuracy at 0.984 and Top-3 at 1.000; 26 tests in GitHub Actions.

InferenceLab | GitHub · Python, vLLM, MLOps

Sep 2026

- Built an async dynamic batcher and TTL/LRU semantic cache behind a swappable vLLM/SGLang/Ollama backend; an 80-request, 16-concurrency workload collapsed into 2 backend batches at 80% cache-hit rate and ~79 ms P95.

SecureRAG | GitHub · FastAPI, RAG, ACL

Sep 2026

- Implemented multi-tenant hybrid retrieval that applies tenant/ACL filters before ranking and repeats live permission checks before context assembly, failing closed on revocation; added Recall@K, MRR, and authorization-leakage tests.

LEADERSHIP

Students' Union, University of Nottingham

Deputy Director

Ningbo, China
May 2023 – Apr 2024

- Organized a multilingual cultural season and film festival (400+ participants) and partnered with KPMG, NIO, Xiaomi, PwC, ByteDance, and Lufthansa on career sessions, each engaging 200+ students.

School of Computer Science, University of Nottingham

Peer Mentor

Ningbo, China
Sep 2023 – Jan 2024

- Mentored 20+ first-years in weekly Java, recursion, data structures, complexity, and Git sessions; authored review sheets and earned helpful ratings from 90%+ of mentees.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, TypeScript, JavaScript, Go, Swift

ML / DL: PyTorch, Hugging Face Transformers, scikit-learn, OpenCV, computer vision, NLP, LoRA, RAG, ONNX, TensorRT

Systems: FastAPI, Spring Boot, Docker, Kubernetes, MySQL, MLflow, vLLM, GitHub Actions, AWS, GCP, Linux